INCH-POUND

MIL-DTL-83723/81F
w/AMENDMENT 1
8 March 2024
SUPERSEDING
MIL-DTL-83723/81F
1 October 2010

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, (CIRCULAR, ENVIRONMENT RESISTING), RECEPTACLE, (SINGLE HOLE MOUNT, BAYONET COUPLING, SOLDER PIN CONTACT), (SERIES III, CLASSES H, J ,P AND Y)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83723.

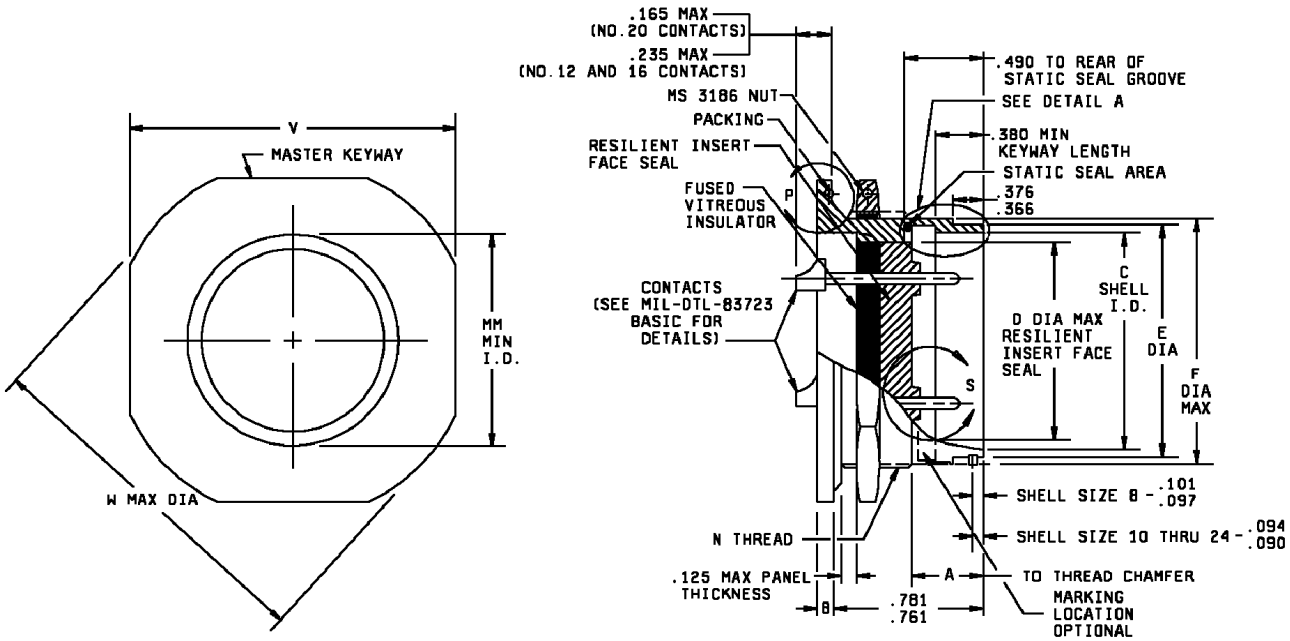


FIGURE 1. Dimensions and configurations.

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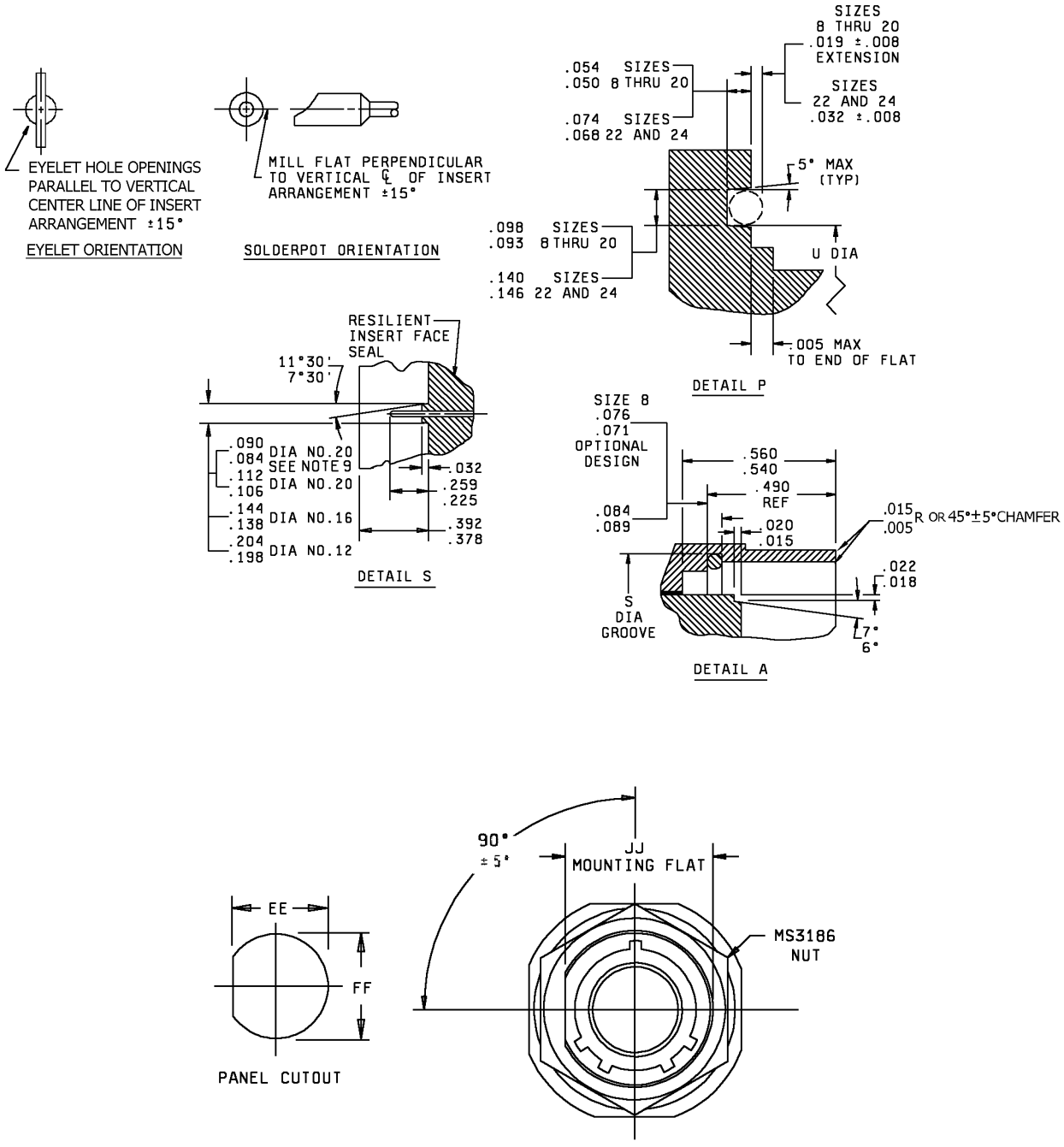


FIGURE 1. Dimensions and configurations - Continued.

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Shell size	A	B	C	D	E	F	N thread
08	.481 .461	.137 .097	.428 .433	.312	.536 .531	.561	.625-20 UN-2A
10	.481 .461	.137 .097	.530 .535	.388	.659 .654	.696	.750-20 UNEF-2A
12	.481 .461	.137 .097	.700 .705	.558	.829 .824	.875	.9375-20 UNEF-2A
14	.481 .461	.137 .097	.769 .774	.627	.898 .893	.935	1.000-20 UNEF-2A
16	.481 .461	.137 .097	.896 .901	.772	1.025 1.020	1.062	1.125-20 UN-2A
18	.481 .461	.137 .097	1.002 1.007	.860	1.131 1.126	1.187	1.250-18 UNEF-2A
20	.481 .461	.137 .097	1.127 1.132	.985	1.256 1.251	1.312	1.375-18 UNEF-2A
22	.481 .461	.168 .128	1.252 1.257	1.110	1.381 1.376	1.437	1.500-18 UNEF-2A
24	.481 .461	.168 .128	1.377 1.382	1.235	1.506 1.501	1.562	1.625-18 UNEF-2A

FIGURE 1. Dimensions and configurations - Continued.

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Shell size	U	V	W	S + .005 - .000	EE	FF	MM	JJ
08	.670	.984 .974	1.068	.508	.605	.635	.493	.596 .590
10	.777	1.109 1.099	1.192	.636	.730	.760	.555	.721 .715
12	.965	1.296 1.286	1.380	.806	.917	.947	.743	.908 .902
14	1.090	1.396 1.386	1.505	.875	.980	1.010	.805	.971 .965
16	1.215	1.521 1.511	1.630	1.002	1.105	1.135	.931	1.096 1.090
18	1.340	1.646 1.636	1.765	1.108	1.229	1.260	1.055	1.220 1.214
20	1.465	1.771 1.761	1.860	1.233	1.354	1.385	1.243	1.345 1.339
22	1.574	1.959 1.949	2.068	1.358	1.479	1.510	1.305	1.470 1.464
24	1.699	2.084 2.074	2.160	1.483	1.604	1.635	1.493	1.595 1.589

FIGURE 1. Dimensions and configurations - Continued.

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NOTES: (Notes 4 through 8 apply to contacts only).

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 inches (0.13 mm) for three place decimals, ± .01 inches (0.25 mm) for two place decimals and ± 1° for all angles.
4. Dimensions apply after plating.
5. Surface finish: 63 **(** $\sqrt{63}$ **)** all over in accordance with ASME-B46.1, unless otherwise specified.
6. Remove all burrs and sharp edges .003 to .005 inches (0.076 to 0.127 mm) R.
7. Diameters to be concentric within .004 inches (0.102 mm) TIR.
8. Diameter in the same cross-sectional plane to be concentric with each other within .003 inches (0.076 mm) TIR.
9. See table I for contact cavities requiring this dimension.

FIGURE 1. Dimensions and configurations - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

This connector mates with MIL-DTL-83723/75 and /77.

For insert arrangements, see MIL-STD-1554 and table I.

TABLE I. Insert arrangements requiring reduced diameters for raised seal barriers on contact cavities as indicated.

Shell size	Insert arrangements	Contact cavities
08	03, 98	1, 2, and 3
10	06	1, 2, 3, 4, 5, and 6

Connector accessories: See SAE-AS85049.

O-ring: See MIL-DTL-83723.

Part or Identifying Number (PIN) example:

M83723/ 81 H 08 55 N

M83723/	Military designation
81	Specification sheet number
H	Class
08	Shell size
55	Insert arrangement
N	Polarizing position

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-83723, this document references the following:

MIL-DTL-83723/75
MIL-DTL-83723/77
MIL-STD-1554
MS3186
ASME-B46.1
SAE-AS85049

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force – 85
DLA - CC

Review activities:
Army - AR, AV, MI
Navy - EC
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5935-2024-060)

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